

Stage 7R Certification and Persistence Boundary Specification

Scientific Results, Derived Indices, Search Workspaces, and Verification Certificates

mdstats

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1 Objective

Stage 7R consolidates the implemented topology, ring, symmetry, and strength stack before Euclidean embedding. It enforces the architectural distinction

scientific result $\neq$ derived index $\neq$ search workspace $\neq$ verification certificate

Runtime/API target:

mdstats 0.19.19a0

No primitive-ring classification, periodic automorphism, symmetry-group order, or bounded strength theorem is changed.

2 Boundary table

Category	Persistent?	Reconstructible?	Example
Scientific result	yes	not assumed	<code>PeriodicNetSymmetry</code> , <code>RingStrengthResult</code>
Derived index	optional	yes	<code>PrimitiveRingSymmetryIndex</code> , <code>PrimitiveRingIndex</code>
Search workspace	no	yes	<code>RingStrengthSearchWorkspace</code>
Positive certificate	yes	independently checkable	<code>RingStrengthWitness</code>
Negative bounded certificate	compact result + deterministic replay	yes	<code>STRONG_IN_DOMAIN</code>

3 Implemented refactors

3.1 Core net symmetry

`PeriodicNetSymmetry` schema v3 stores only:

- source view/topology identities;
- normalized automorphism representatives;
- multiplication table;
- translation cocycle;
- inverse and identity indices;
- vertex orbits; and
- edge orbits.

It contains no primitive-ring keys, actions, or ring-search provenance.

3.2 Primitive-ring symmetry index

`PrimitiveRingSymmetryIndex` binds one exact group to one exact ring catalog and uses integer target positions in the action table. It records ring-catalog completeness metadata and validates the induced action homomorphism.

3.3 Strength result versus workspace

`RingStrengthResult` schema v2 stores:

```
target_placement
domain
resources
status
diagnostics
candidate_set_digest
witness | None
```

It no longer stores every candidate placement. The deterministic transient `RingStrengthSearchWorkspace` stores the exhaustive candidate tuple when a caller explicitly asks for it.

For a candidate set $\mathcal{P}$, the persistent record stores

$$d_{\mathcal{P}} = \text{SHA256}(\text{canonical_json}(\mathcal{P})).$$

3.4 Independent strength verification

`RingStrengthResult.verify(index)` performs two checks:

1. for `WEAK_CERTIFIED`, recompute the physical lifted-edge parity of the target and witness components;
2. deterministically rebuild the declared finite domain, rerun the bounded cancellation solve, and require canonical result equality.

`from_dict(..., verify=True)` uses this path by default. Recomputing a payload digest is therefore insufficient to forge a scientific certificate.

3.5 GF(2) memory policy

The finite cancellation backend now declares conservative capacities:

```
FiniteRingCancellationResources(
    max_matrix_bits,
    max_provenance_bits,
)
```

For N candidates and E physical edge basis elements, the preflight estimates

$$B_{\text{matrix}} = NE, \quad B_{\text{provenance}} = N \min(N, E).$$

Exceeding either bound raises a resource error. The strength layer translates it to `UNRESOLVED_TRUNCATED`; it never reports a false negative strength result.

3.6 Shared barycentric placement

The exact rational equilibrium solve is extracted to `PeriodicBarycentricPlacement`. Symmetry discovery consumes that source-bound object instead of carrying an anonymous private coordinate tuple.

3.7 Fast view lookup

`PeriodicNetView` constructs transient immutable maps

atom index -> vertex position

FrameworkEdgeKey -> edge position

so repeated automorphism and ring-action lookups are $O(1)$ rather than linear searches.

4 Schema changes

Object	Previous	Current
<code>PeriodicNetSymmetry</code>	v2	v3
<code>PeriodicNetSymmetryDiscovery</code>	v1	v2
<code>RingStrengthResult</code>	v1	v2
<code>RingStrengthCatalog</code>	v1	v2
<code>PrimitiveRingSymmetryIndex</code>	absent	v1
<code>PeriodicBarycentricPlacement</code>	absent	v1

These are intentional alpha API changes. Legacy payloads are not silently upgraded because their ownership boundaries are ambiguous.

5 Measured Na-LTA effect

For the 96-operation Na-LTA symmetry and 82 primitive rings:

```
core PeriodicNetSymmetry JSON      ~0.83 MB
PrimitiveRingSymmetryIndex JSON    ~1.01 MB
combined discovery JSON            ~1.84 MB
```

The former combined ring-heavy symmetry payload was approximately 9.17 MB.

For one depth-eight 8-ring strength calculation with 3,240 candidate placements:

`persistent RingStrengthResult JSON ~4 KB`
`candidate workspace transient/reconstructible`

The earlier persistent result scaled approximately linearly with every complete candidate key and placement.

6 Deferred work

Stage 7R deliberately does not implement:

- symmetry-orbit-level strength transport;
- a sparse or two-pass $\text{GF}(2)$ solver;
- a compact standalone negative row-space certificate;
- a Euclidean invariant Gram matrix;
- automatic disk/face construction; or
- natural tiling.

The symmetry-equivariant strength layer remains required before final face selection, but it does not block Stage 8A embedding.

7 Acceptance gate

The consolidation passes when:

1. core symmetry round trips without ring data;
2. ring action round trips only with the exact group and ring catalog;
3. strength payloads contain no candidate workspace;
4. forged weak witnesses fail independent verification;
5. $\text{GF}(2)$ memory limits return unresolved status;
6. direct barycentric placement and discovery produce identical exact coordinates and symmetry groups;
7. all prior Stage-4–7 scientific regressions pass; and
8. package-wide tests reveal no import or schema regressions.